

to 35% higher than in Taiwan, due to labour, energy, and permitting delays. India faces similar issues—land acquisition hurdles, regulatory bottlenecks, and underdeveloped infrastructure contribute to elevated project risks and delays.

Material sourcing presents further complications. Advanced chips rely on rare elements like germanium, tungsten, and gallium, most of which are controlled by a few countries—China, for instance, accounts for over 80% of the global gallium supply. India must secure strategic imports and diversify supply chains, despite lacking local reserves and robust trade partnerships.

Advanced packaging is another weak link. Over 90% of global capacity is concentrated in East Asia, dominated by players such as ASE and TSMC. Talent, too, is a pressing concern. While India produces strong engineering graduates, few are specifically trained in semiconductor manufacturing.

For perspective, the US anticipates a shortfall of 70,000-90,000 skilled workers in the sector by 2030, even with reinforced training pipelines. India will need to address this workforce gap decisively if it aims to turn its semiconductor vision into reality.

Where is India then...

“India’s government incentives are quite competitive, especially for trailing-edge and compound semiconductor fabs,” says industry and policy veteran Pranay Kotasthane, Deputy Director of the Takshashila Institution.

India’s semiconductor policies, particularly the ISM, address challenges in cost, supply chain development, packaging, and talent acquisition. The ISM offers up to 50% fiscal support for approved semiconductor fabrication models.

State governments such as Gujarat, Odisha, Madhya Pradesh, Andhra Pradesh, Tamil Nadu, Karnataka, Telangana and others supplement this with additional incentives, bringing total subsidies to approximately 70-75% of model costs. This makes India’s subsidy regime among the most generous globally.

The government has recognised the need to develop semiconductor-grade raw materials, including silicon wafers, high-purity gases, and speciality chemicals. In Budget 2025-26, customs duties were eliminated on waste and scrap of twelve critical minerals, including tungsten—a key material used in chipmaking. While not all are directly tied to fabrication, this step reflects a broader effort to ease access to strategic inputs and reduce dependence on imports.

To address the talent gap in semiconductor design and manufacturing, targeted initiatives such as the Design Linked Incentive (DLI) scheme and Chips to Startup (C2S) provide financial support to startups and academic institutions. For instance, Bengaluru-based Mindgrove Technologies is developing low-power chips for IoT applications under this scheme, while IIT Bombay has received funding to establish a VLSI design and training hub.

Experts believe that in contrast to the uncertain economics of fabrication, India’s chip design and talent efforts are showing clear, scalable progress, laying the foundation for sustained momentum in a turbulent global tech landscape.

According to Dr Satya Gupta, President of the VLSI Society of India and a key figure in India’s semiconductor strategy, building a globally competitive ecosystem demands a triad approach: “For India to become a strong Semiconductor Product Nation, we need to

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